# Introduction to Algorithms and Data Structures

week 1: Arrays Revisited

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# Discussing homework from the last lesson

- Where you able to do the exercises?
  1. Yes! All exercises are done!
  2. No.., Because I have these questions
  3. I haven't prioritized this subject enough at the moment to do the assignments

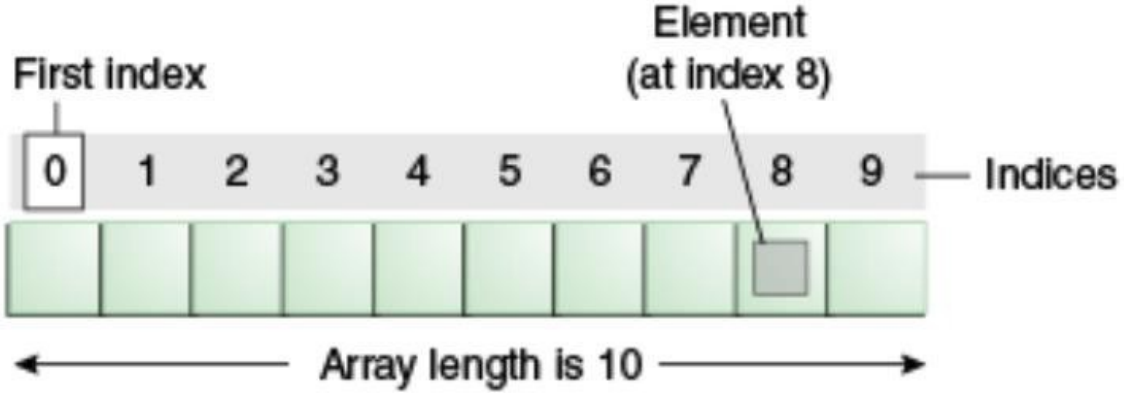
# Arrays Revisited

- **What is an Array?**
- Similarities and differences between an Array and a ArrayList in java
- How to solve problems by using arrays
- homework

## What is an Array?

- A datatype representing a collection of items
- Each item within an array is of the same type
- The array keeps track where each item is stored in memory
- Storing and retrieving items is based on index
- Has a fixed length
- Iterable: you can loop over the contents

```
int[] myIntArray = new int[10];
```



## How do we use an array?

- Create a new array: empty: `int[] myNumbers = new int[3];`
- pre-filled: `int[] myNumbers = {2,4,6};`

- Access an element: `myNumbers[2];`

- Overwrite an element: `myNumbers[1] = 100;`

- Delete an element: ???

# What is an Array?

```
char[] someArray = {'U','B','F','D','A','E','C'};
```

1. What is the length of this array?
2. What type do we store in this array?
3. What's the value of: someArray[4]?
4. What's the value of: someArray[7]?

Are the following assignments allowed?

1. someArray[4] = *'K'*;
2. someArray[4] = "something";
3. someArray[4] = someArray[2];

| U | B | F | D | A | E | C |
|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 |

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# Similarities & Differences between Array and ArrayList

| Similarities | Arrays | ArrayLists |
|---|---|---|
| Order of items | Order of insertion | |
| Iterable | By (enhanced for) loops (iterators) | |
| Accessing elements | By index | |

| Differences | | Arrays | ArrayLists |
|---|---|---|---|
| Size | | Fixed (on creation) | Dynamic |
| Type of elements | Primitive | Yes | *No** |
| | Reference | Yes | Yes |
| 'Direct Access' | | Yes | No (through add/get()) |
| Selling point | | *Faster*** | Flexibility |

**Yes, through autoboxing*
***Depends on the situation*

## What to use?

Keeping track of how many times a certain roll occurs when throwing a dice?

# What to use?

Storing all the students in a university?

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# What to use?

Keeping track of all the effects that are applied to a character in a game?

## What to use?

Creating a data-model that stores all possible locations on a gameboard?

## Workshop:

Do exercise: 1_arrayBasics from **arrays**

*You can find this on Blackboard > week1*

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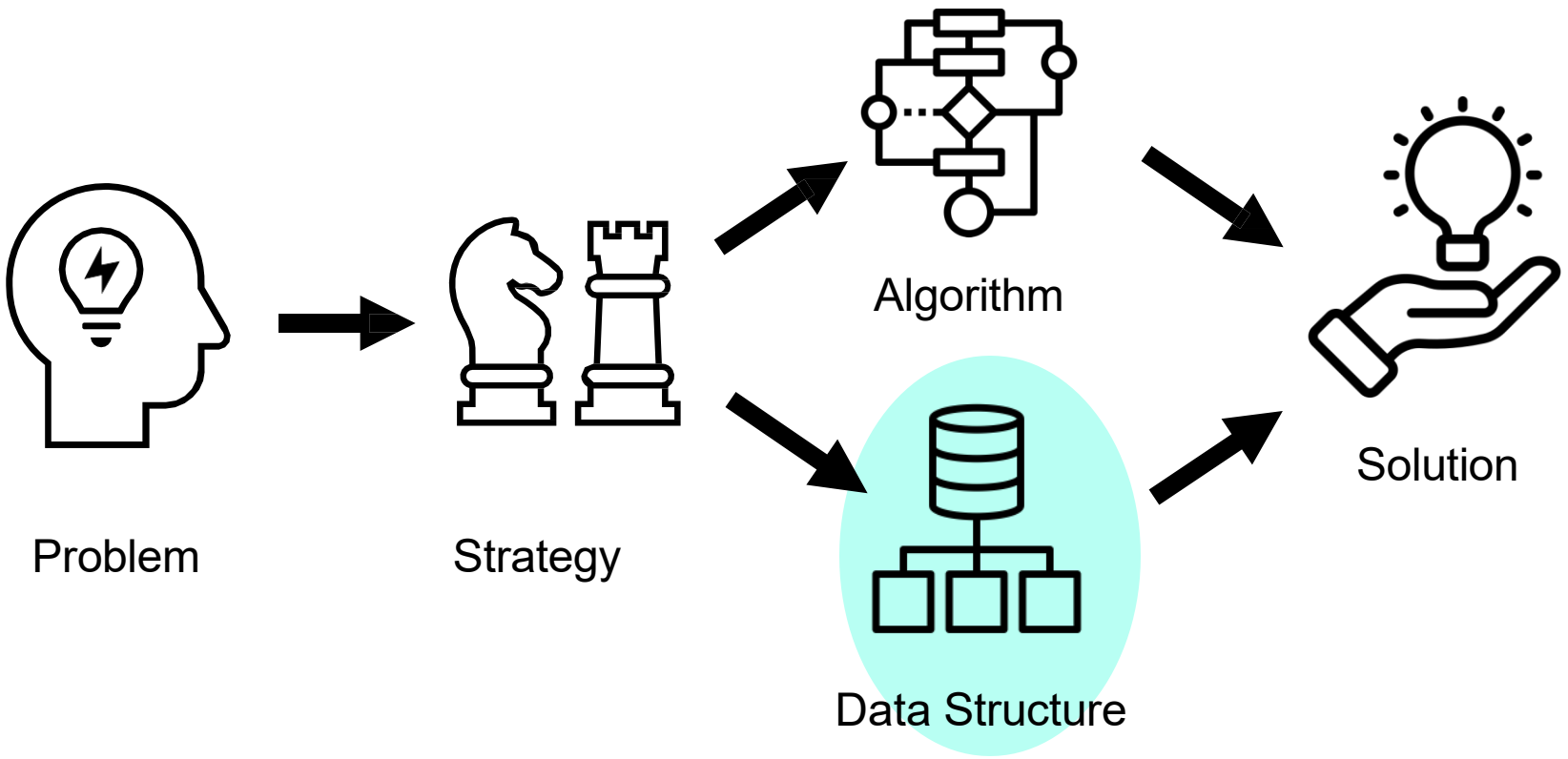
# Problem Solving – Algorithm first or data structure first?



## Workshop:

Do exercise: 2_cipher from **arrays**

*You can find this on Blackboard > week1*

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# Now it’s your turn!